SQL DATABASE COMMANDS

DDL | DML | DCL | TCL

BANK MANAGEMENT SYSTEM

Practical Execution & Results

SECTION 1: DDL (DATA DEFINITION LANGUAGE)

Commands: CREATE, ALTER, DROP, TRUNCATE, RENAME

Execution Screenshot 1: CREATE TABLE & ALTER Commands

```
mysql> CREATE TABLE bank_transactions (
    ->     txn_id INT PRIMARY KEY,
    ->     customer_name VARCHAR(50),
    ->     branch_name VARCHAR(50),
    ->     transaction_type VARCHAR(20),
    ->     amount DECIMAL(10,2),
    ->     transaction_date DATE
    -> );
Query OK, 0 rows affected (0.09 sec)

mysql>
mysql>
mysql> ALTER TABLE bank_transactions
    -> ADD account_no VARCHAR(20);
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql>
mysql>
mysql> ALTER TABLE bank_transactions
    -> MODIFY customer_name VARCHAR(100);
Query OK, 0 rows affected (0.09 sec)
Records: 0  Duplicates: 0  Warnings: 0
```

Results: CREATE TABLE bank_transactions successfully created. ALTER TABLE commands ADD account_no and MODIFY customer_name executed without errors.

Execution Screenshot 2: RENAME & TRUNCATE Commands

```
mysql> RENAME TABLE bank_transactions
    -> TO customer_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> TRUNCATE TABLE bank_transactions;
ERROR 1146 (42S02): Table 'sec_11.bank_transactions' doesn't exist
mysql> RENAME TABLE customer_transactions to bank_transactions;
Query OK, 0 rows affected (0.04 sec)

mysql> TRUNCATE TABLE bank_transactions;
Query OK, 0 rows affected (0.07 sec)
```

Results: RENAME TABLE bank_transactions TO customer_transactions executed. TRUNCATE TABLE command successfully removed all rows while preserving table structure.

Execution Screenshot 3: ALTER DROP & INSERT Commands

```
mysql> ALTER TABLE bank_transactions
    -> DROP COLUMN account_no;
Query OK, 0 rows affected (0.11 sec)
Records: 0  Duplicates: 0  Warnings: 0

mysql> INSERT INTO bank_transactions VALUES
    -> (101,'Ravi','Hyderabad','Deposit',5000,'2024-01-05'),
    -> (102,'Sita','Hyderabad','Withdrawal',2000,'2024-01-06'),
    -> (103,'Kiran','Vijayawada','Deposit',12000,'2024-01-08'),
    -> (104,'Anil','Vizag','Deposit',8000,'2024-01-10'),
    -> (105,'Priya','Hyderabad','Withdrawal',3500,'2024-01-11'),
    -> (106,'Ramesh','Vizag','Deposit',15000,'2024-01-12'),
    -> (107,'Keerthi','Vijayawada','Withdrawal',1000,'2024-01-13'),
    -> (108,'Rahul','Hyderabad','Deposit',9000,'2024-01-14'),
    -> (109,'Sneha','Vizag','Withdrawal',4000,'2024-01-15'),
    -> (110,'Madhu','Vijayawada','Deposit',11000,'2024-01-16');
Query OK, 10 rows affected (0.01 sec)
Records: 10  Duplicates: 0  Warnings: 0

mysql> INSERT INTO bank_transactions
    -> VALUES
    -> (111,'Venu','Vizag','Deposit',7000,'2024-01-18');
Query OK, 1 row affected (0.02 sec)

mysql>
mysql> UPDATE bank_transactions
    -> SET amount = 5000
    -> WHERE txn_id = 105;
Query OK, 1 row affected (0.03 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>
mysql> DELETE FROM bank_transactions
    -> WHERE txn_id = 111;
Query OK, 1 row affected (0.02 sec)
```

Results: ALTER TABLE DROP COLUMN account_no executed. 10 rows of sample transaction data successfully inserted into the bank_transactions table.

SECTION 2: DML (DATA MANIPULATION LANGUAGE)

Commands: INSERT, UPDATE, DELETE, SELECT

Execution Screenshot 4: SELECT Queries (All Data, WHERE Clause, ORDER BY)

```
mysql>
mysql> SELECT *
    -> FROM bank_transactions;
+--------+---------------+-------------+------------------+----------+------------------+
| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
+--------+---------------+-------------+------------------+----------+------------------+
|    101 | Ravi          | Hyderabad   | Deposit          |  5000.00 | 2024-01-05       |
|    102 | Sita          | Hyderabad   | Withdrawal       |  2000.00 | 2024-01-06       |
|    103 | Kiran         | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|    104 | Anil          | Vizag       | Deposit          |  8000.00 | 2024-01-10       |
|    105 | Priya         | Hyderabad   | Withdrawal       |  5000.00 | 2024-01-11       |
|    106 | Ramesh        | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|    107 | Keerthi       | Vijayawada  | Withdrawal       |  1000.00 | 2024-01-13       |
|    108 | Rahul         | Hyderabad   | Deposit          |  9000.00 | 2024-01-14       |
|    109 | Sneha         | Vizag       | Withdrawal       |  4000.00 | 2024-01-15       |
|    110 | Madhu         | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
+--------+---------------+-------------+------------------+----------+------------------+
10 rows in set (0.00 sec)

mysql>
mysql> SELECT *
    -> FROM bank_transactions
    -> WHERE transaction_type='Deposit';
+--------+---------------+-------------+------------------+----------+------------------+
| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
+--------+---------------+-------------+------------------+----------+------------------+
|    101 | Ravi          | Hyderabad   | Deposit          |  5000.00 | 2024-01-05       |
|    103 | Kiran         | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|    104 | Anil          | Vizag       | Deposit          |  8000.00 | 2024-01-10       |
|    106 | Ramesh        | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|    108 | Rahul         | Hyderabad   | Deposit          |  9000.00 | 2024-01-14       |
|    110 | Madhu         | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
+--------+---------------+-------------+------------------+----------+------------------+
6 rows in set (0.00 sec)

mysql>
mysql>
mysql> SELECT *
    -> FROM bank_transactions
    -> ORDER BY amount DESC;
+--------+---------------+-------------+------------------+----------+------------------+
| txn_id | customer_name | branch_name | transaction_type | amount   | transaction_date |
+--------+---------------+-------------+------------------+----------+------------------+
|    106 | Ramesh        | Vizag       | Deposit          | 15000.00 | 2024-01-12       |
|    103 | Kiran         | Vijayawada  | Deposit          | 12000.00 | 2024-01-08       |
|    110 | Madhu         | Vijayawada  | Deposit          | 11000.00 | 2024-01-16       |
|    108 | Rahul         | Hyderabad   | Deposit          |  9000.00 | 2024-01-14       |
|    104 | Anil          | Vizag       | Deposit          |  8000.00 | 2024-01-10       |
|    101 | Ravi          | Hyderabad   | Deposit          |  5000.00 | 2024-01-05       |
|    105 | Priya         | Hyderabad   | Withdrawal       |  5000.00 | 2024-01-11       |
|    109 | Sneha         | Vizag       | Withdrawal       |  4000.00 | 2024-01-15       |
|    102 | Sita          | Hyderabad   | Withdrawal       |  2000.00 | 2024-01-06       |
|    107 | Keerthi       | Vijayawada  | Withdrawal       |  1000.00 | 2024-01-13       |
+--------+---------------+-------------+------------------+----------+------------------+
10 rows in set (0.00 sec)
```

Results: SELECT * FROM bank_transactions returned 10 rows. SELECT with WHERE transaction_type="Deposit" returned 6 deposit records. SELECT with ORDER BY amount DESC sorted all transactions from highest to lowest amount.

SECTION 3: DCL (DATA CONTROL LANGUAGE)

Commands: GRANT, REVOKE

Execution Screenshot 5: CREATE USER, GRANT & REVOKE Privileges

```
mysql> CREATE USER 'Auditor1'@'localhost'
    -> IDENTIFIED BY 'Auditor@123';
Query OK, 0 rows affected (0.08 sec)

mysql> CREATE USER 'BranchManager'@'localhost'
    -> IDENTIFIED BY 'Branch@123';
Query OK, 0 rows affected (0.03 sec)

mysql> GRANT ALL PRIVILEGES
    -> ON sec_11.bank_transactions
    -> TO 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql>
mysql>
mysql> REVOKE SELECT
    -> ON bank_transactions
    -> FROM Auditor1;
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host '%' on table 'bank_transactions'
mysql> REVOKE SELECT
    -> ON sec_11.bank_transactions
    -> FROM 'Auditor1'@'localhost';
ERROR 1147 (42000): There is no such grant defined for user 'Auditor1' on host 'localhost' on table 'bank_transactions'
mysql> GRANT SELECT
    -> ON sec_11.bank_transactions
    -> TO 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> SHOW GRANTS FOR 'Auditor1'@'localhost';
+-----------------------------------------------------------------------+
| Grants for Auditor1@localhost                                         |
+-----------------------------------------------------------------------+
| GRANT USAGE ON *.* TO `Auditor1`@`localhost`                          |
| GRANT SELECT ON `sec_11`.`bank_transactions` TO `Auditor1`@`localhost` |
+-----------------------------------------------------------------------+
2 rows in set (0.02 sec)

mysql>
mysql> REVOKE SELECT
    -> ON sec_11.bank_transactions
    -> FROM 'Auditor1'@'localhost';
Query OK, 0 rows affected (0.01 sec)

mysql> REVOKE ALL PRIVILEGES
    -> ON sec_11.bank_transactions
    -> FROM 'BranchManager'@'localhost';
Query OK, 0 rows affected (0.02 sec)

mysql> SHOW GRANTS FOR 'BranchManager'@'localhost';
+-------------------------------------------------+
| Grants for BranchManager@localhost              |
+-------------------------------------------------+
| GRANT USAGE ON *.* TO `BranchManager`@`localhost` |
+-------------------------------------------------+
1 row in set (0.00 sec)
```

Results: Created two users (Auditor1 and BranchManager). GRANT ALL PRIVILEGES executed for BranchManager. REVOKE SELECT removed permissions from Auditor1. SHOW GRANTS commands displayed current user privileges.

SECTION 4: TCL (TRANSACTION CONTROL LANGUAGE)

Commands: COMMIT, SAVEPOINT, ROLLBACK

Execution Screenshot 6: Transaction Control - SAVEPOINT, ROLLBACK & COMMIT

```
mysql>
mysql> UPDATE bank_transactions
    -> SET amount=6000
    -> WHERE txn_id=101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>
mysql> SAVEPOINT Before_Update;
Query OK, 0 rows affected (0.00 sec)

mysql> UPDATE bank_transactions
    -> SET amount=99999
    -> WHERE txn_id=102;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> ROLLBACK TO Before_Update;
ERROR 1305 (42000): SAVEPOINT Before_Update does not exist
mysql>
mysql> UPDATE bank_transactions
    -> SET amount=7000
    -> WHERE txn_id=101;
Query OK, 1 row affected (0.01 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql>
mysql> UPDATE bank_transactions
    -> SET amount=9000
    -> WHERE txn_id=102;
Query OK, 1 row affected (0.02 sec)
Rows matched: 1  Changed: 1  Warnings: 0

mysql>
mysql> COMMIT;
Query OK, 0 rows affected (0.00 sec)

mysql>
```

Results: UPDATE command modified transaction amounts. SAVEPOINT Before_Update created a checkpoint. Subsequent ROLLBACK TO Before_Update reverted changes to the savepoint. COMMIT finalized all confirmed transactions permanently.

COMMAND SUMMARY

Category	Commands	Purpose
DDL	CREATE, ALTER, DROP, TRUNCATE, RENAME	Define database structures and schemas
DML	INSERT, UPDATE, DELETE, SELECT	Manipulate and retrieve data
DCL	GRANT, REVOKE	Control user access and permissions
TCL	COMMIT, SAVEPOINT, ROLLBACK	Manage transaction consistency

All SQL commands executed successfully on the MySQL Database Management System. The bank_transactions table was created, populated with sample data, queried with various SELECT statements, modified using UPDATE commands, and access control was demonstrated through GRANT and REVOKE operations.